



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.NOS.1

Divide Whole Numbers by Fractions

Lesson 6-9 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can divide a whole number by a fraction by multiplying by the reciprocal.

Language Objective: I can explain my steps using the words fraction, reciprocal, quotient, and Keep, Change, Flip.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Whole Number

Fraction

Reciprocal

Keep, Change, Flip

Quotient



Tap any word to see its meaning and a picture.

1 A counting number like 0, 1, 2, or 3 with no fraction or decimal part is a

2 A number that shows part of a whole, like $\frac{3}{4}$, is a

3 The fraction you get by flipping a fraction upside down is its

4 To divide by a fraction, I : keep the first number, change $\div$ to $\times$, and flip the second fraction.

5 The answer to a division problem is the

Write About the Math

How does dividing a whole number by a fraction help the detective?

¿Cómo ayuda dividir un número entero entre una fracción al detective?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Whole Number**
Número entero

☐ **Fraction**
Fracción

☐ **Reciprocal**
Recíproco

☐ **Keep, Change, Flip**
*Mantener, cambiar,
invertir*

☐ **Quotient**
Cociente

Model: *Each whole pound makes 2 half-pound bags, so 4 pounds makes 8 bags.* — Students see that $4 \div 1/2 = 8$ and explain there are 2 halves in each whole pound.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can set up ____ markers because $10 \div 1/4 =$ ____.**
- **My answer is ____ because ____.**

Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
Mi afirmación es ____.
- **My evidence is ____, so ____.**
Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Whole Number
2. **Notes 2:** Fraction
3. **Notes 3:** Reciprocal
4. **Notes 4:** Keep, Change, Flip
5. **Notes 5:** Quotient
6. **Watch for this mistake:** *A common mistake in Divide Whole Numbers by Fractions is doing 'Keep, Change' but skipping the 'Flip' — students multiply by the divisor as it is instead of by its reciprocal. For example, for $6 \div 3/4$, a student might write $6 \times 3/4 = 4\ 1/2$ and stop there, but the divisor $3/4$ must be flipped to $4/3$ first: $6 \times 4/3 = 24/3 = 8$. Before you submit, check: did I flip the second fraction (the divisor) upside down before multiplying?*
7. **Try It 1:** A. $4\ 1/2$ pouches (4 full, half a pouch left) — $3 \div 2/3 = 3 \times 3/2 = 9/2 = 4\ 1/2$. You can fill 4 full pouches with half a pouch of mix left over.
8. **Try It 2:** A. $12 - 9 \div 3/4 = 9 \times 4/3 = 36/3 = 12$ pans.